

Fluorescence

Y25 (2025) — English translation

Equipment (electrical): Photosensitive CCD ruler connected to a computer (only the program for the CCD ruler is used). Magnetic screen USB flash drive lamp

Equipment (electrical):

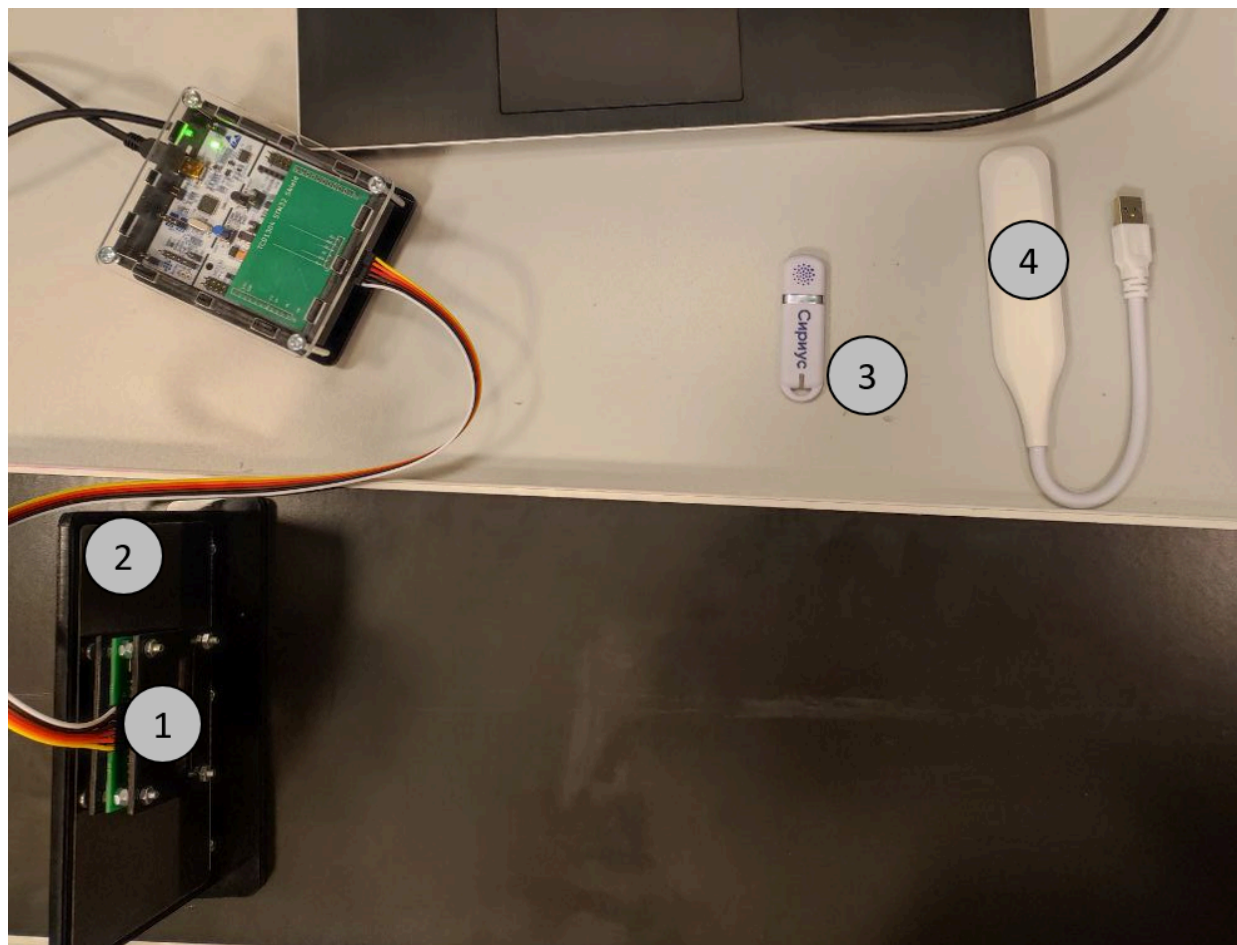


Fig. 1.

Equipment (optical): Wire power supply and cable adapter. Do not change the voltage 15 B. Loop Continuous spectrum LED LED-CON. Three-color LED-RGB LED (colors can be changed using a key on the adapter). Three black lenses L_1 with optical power $+6 \text{ dmp}$. One white lens L_2 with optical power $+16 \text{ dmp}$. Lens L_3 on a rod with unknown optical power (lens L_3 can be mounted directly on a tripod). Diffraction grating with period $d_0 = 2.0 \text{ mkm}$. Optical table. Stand for illuminator and cuvettes. Spacer (required for part C) Stand for lens L_3 and laser. Ruler Slit

Equipment (optical):

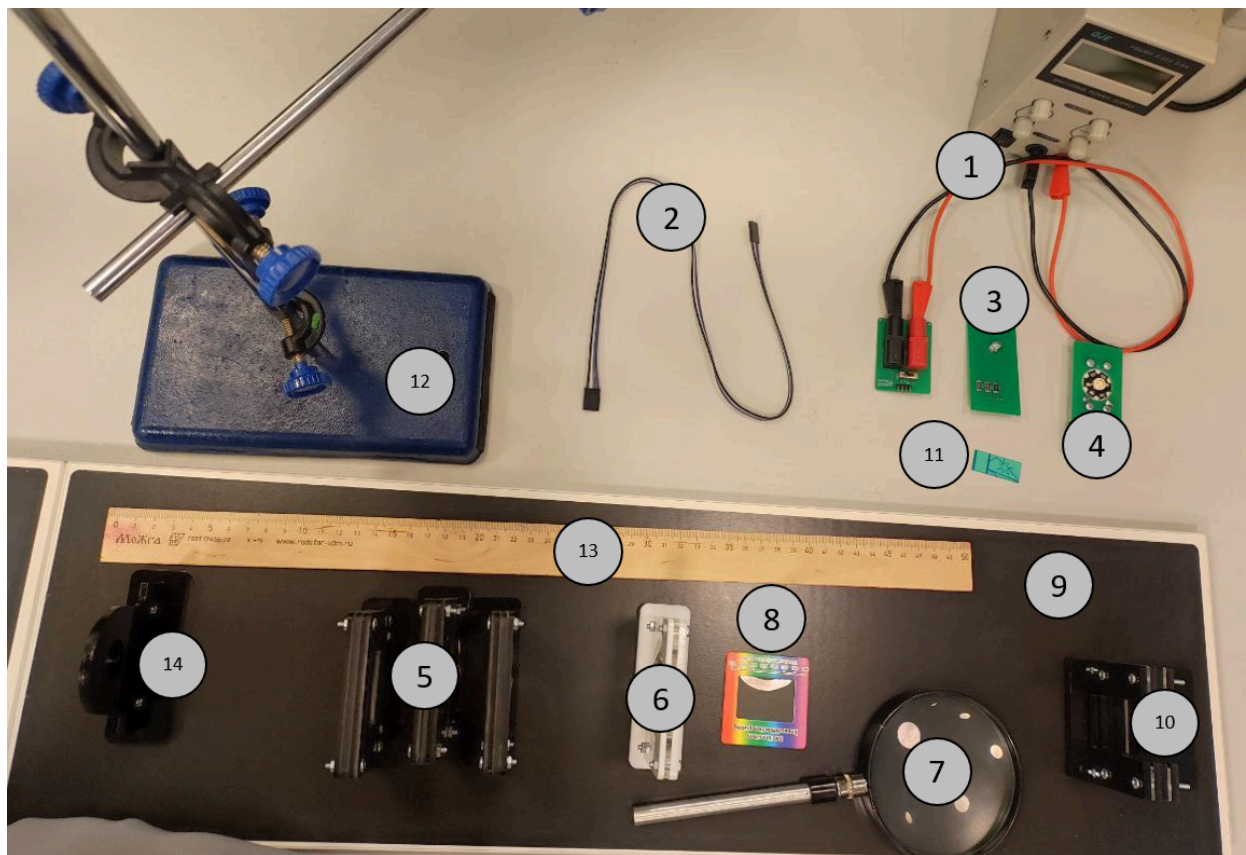


Fig. 2.

Equipment (chemical): Cuvettes with lids (10 pieces) Jars for analysis with water (2 pieces) Syringe with a 0.5 ml needle Two syringes with a 3 ml needle Lasers (red, green and blue) Batteries for lasers (6 pieces) Laser button holder Rhodamine B solution with molar concentration $c_0 = 200 \mu\text{M}$ Tweezers

Equipment (chemical):

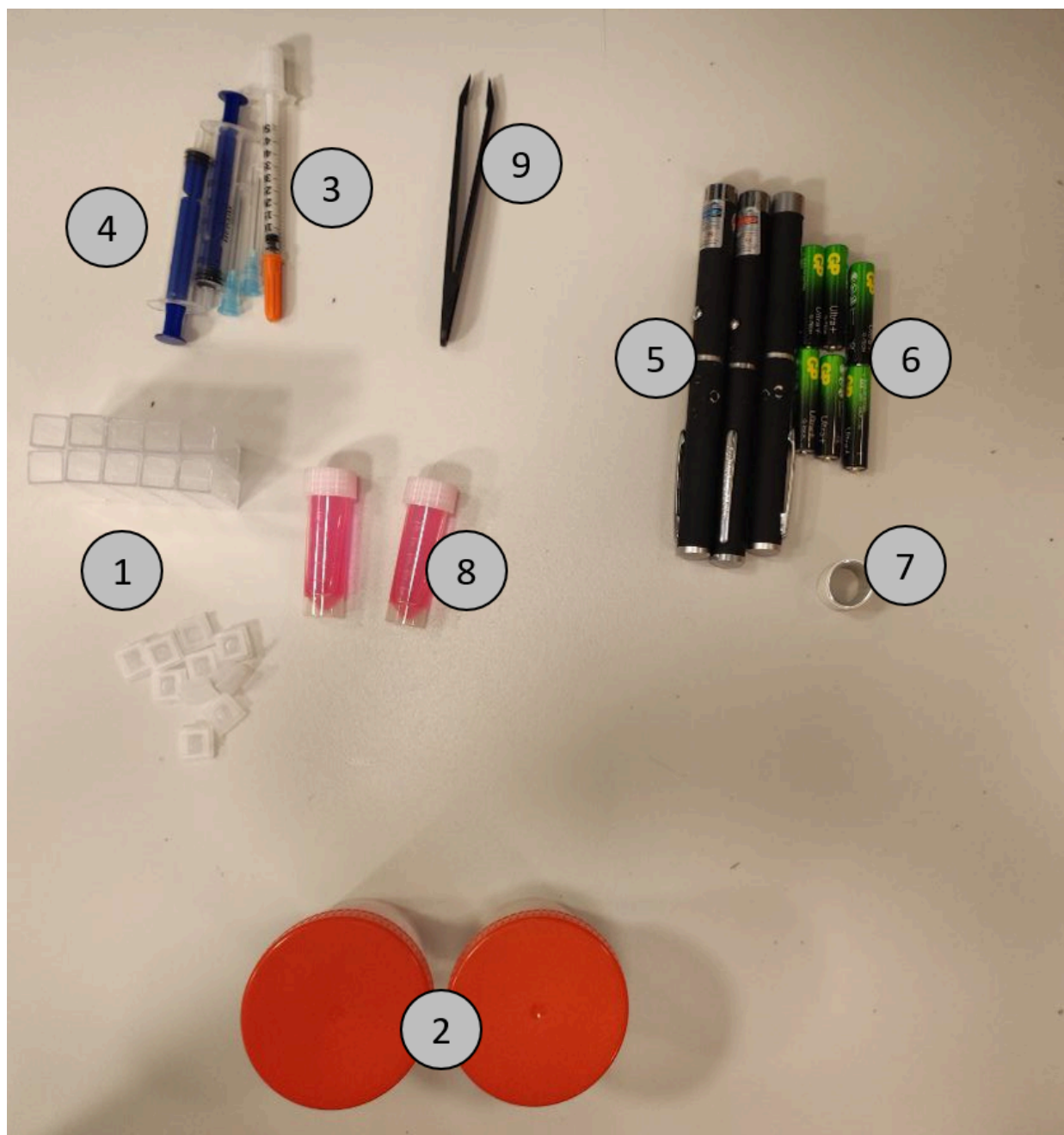


Fig. 3.

ВНИМАНИЕ. ВСЕ ЭКСПЕРИМЕНТАЛЬНЫЕ ДАННЫЕ ДОЛЖНЫ БЫТЬ СОХРАНЕНЫ НА ФЛЕШКЕ В ПАПКЕ С ВАШЕЙ ФАМИЛИЕЙ. НЕПРАВИЛЬНО СОХРАНЕННЫЕ ФАЙЛЫ НЕ ОЦЕНИВАЮТСЯ. Если вы хотите сохранить какие-то дополнительные данные, то сохраните их в папке Add. Такие файлы будут игнорироваться, если к ним не будет сделано описание в работе. На флешке нельзя создавать какие-либо новые папки или удалять существующие. При сохранении файлов с помощью программы spectrometer.bat расширение .csv устанавливается автоматически. Изучение электромагнитных свойств веществ является основой современной науки, так как фотоны являются самой «дешевой» частицей, которую можно отправить взаимодействовать с изучаемым образцом. В данной задаче мы рассмотрим такие техники исследования как спектроскопия поглощения и флуоресцентная микроскопия. Свет как электромагнитная волна взаимодействует с заряженными частицами: электронами и ядрами в молекулах. С точки зрения классической механики полная энергия такой системы является параметром, определяющим (естественно, не полностью) движение системы, причём этот параметр может изменяться непрерывно. Однако с экспериментом согласуется другое, квантовое рассмотрение. В этом случае оказывается, что полная энергия системы

может принимать только некоторый дискретный спектр значений. Каждой энергии соответствует некоторое квантовое состояние системы. Спектр энергий ограничен снизу, то есть существует минимально возможное значение энергии системы. Соответствующее ему состояние называется основным состоянием. Состояния с большей энергией называются возбуждёнными. Для наглядного представления энергетического спектра молекулы используют диаграмму Яблонского

Если вы хотите сохранить какие-то дополнительные данные, то сохраните их в папке Add. Такие файлы будут игнорироваться, если к ним не будет сделано описание в работе. На флешке нельзя создавать какие-либо новые папки или удалять существующие.

При сохранении файлов с помощью программы spectrometer.bat расширение .csv устанавливается автоматически.

Изучение электромагнитных свойств веществ является основой современной науки, так как фотоны являются самой «дешевой» частицей, которую можно отправить взаимодействовать с изучаемым образцом. В данной задаче мы рассмотрим такие техники исследования как спектроскопия поглощения и флюоресцентная микроскопия.

Свет как электромагнитная волна взаимодействует с заряженными частицами: электронами и ядрами в молекулах. С точки зрения классической механики полная энергия такой системы является параметром, определяющим (естественно, не полностью) движение системы, причём этот параметр может изменяться непрерывно.

Однако с экспериментом согласуется другое, квантовое рассмотрение. В этом случае оказывается, что полная энергия системы может принимать только некоторый дискретный спектр значений. Каждой энергии соответствует некоторое квантовое состояние системы. Спектр энергий ограничен снизу, то есть существует минимально возможное значение энергии системы. Соответствующее ему состояние называется основным состоянием. Состояния с большей энергией называются возбуждёнными.

Для наглядного представления энергетического спектра молекулы используют диаграмму Яблонского

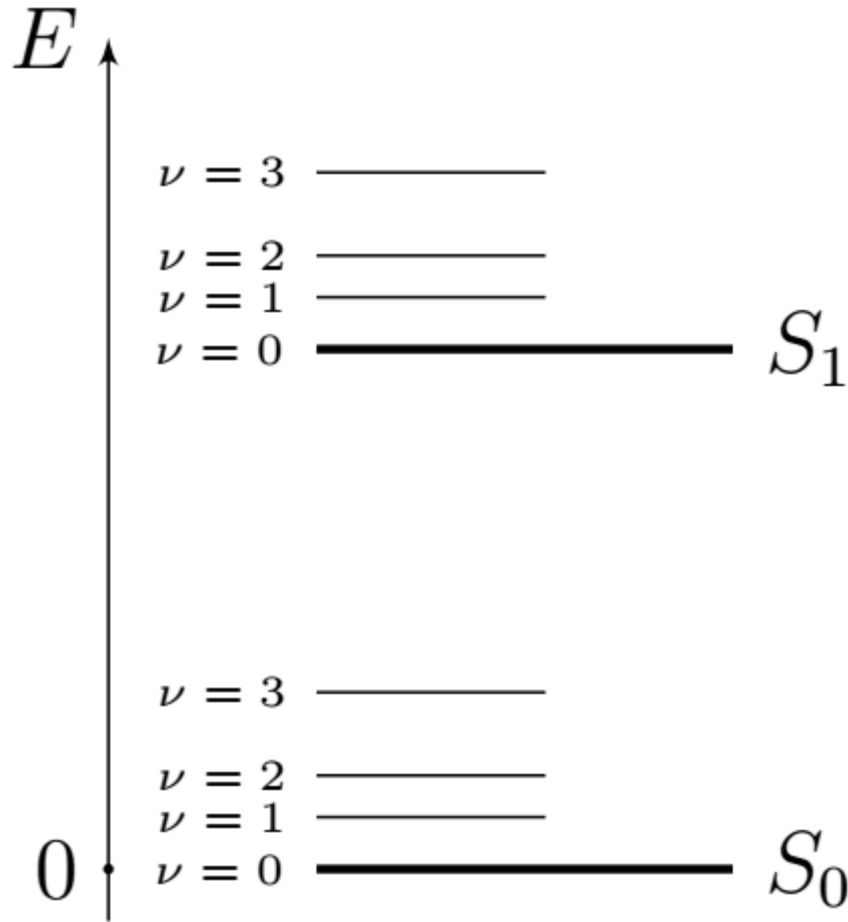


Fig. 4.

In it, the energy of the molecule is plotted vertically and the energy levels are indicated by horizontal lines. In this problem we will consider the interaction of light with Rhodamine B, which we will call RhB. The energy of the molecule RhB E is contributed by the energy of electron motion E_e and the energy of atomic vibrations E_a , while the energy of atomic vibrations turns out to be significantly less than the energy of electron motion. The motion of electrons and vibrations of atoms can be considered independently. Let us denote by S_0 the ground state of the molecule, that is, one in which the energies of electron motion and atomic vibrations are minimal. Since the total energy can be determined up to a constant, we set the energy of this state to zero. By S_1 we denote the state of the molecule in which the electrons are in the first excited state, and the vibrational energy of the atoms is also minimal. The corresponding levels are highlighted in the Jablonski diagram with bold lines. Above each of the electronic levels S_0, S_1 lies a group of levels corresponding to the excited vibrational states of atoms. The number of the vibrational level will be denoted by the letter ν . Summarizing all of the above, the energy of a molecule consists of the energy of electron motion and the vibration energy of atoms. As an example, the energy $E_{S_1, \nu=1}$ of the molecule state $S_1, \nu = 1$ is the sum of the energy of electrons E_{S_1} and the vibrational energy of atoms $E_{\nu=1}$:

$$E_{S_1, \nu=1} = E_{S_1} + E_{\nu=1}$$

All photon absorption processes are possible when the molecule is in the $S_0, \nu = 0$ state (left), in the $S_0, \nu = 2$ state (right).

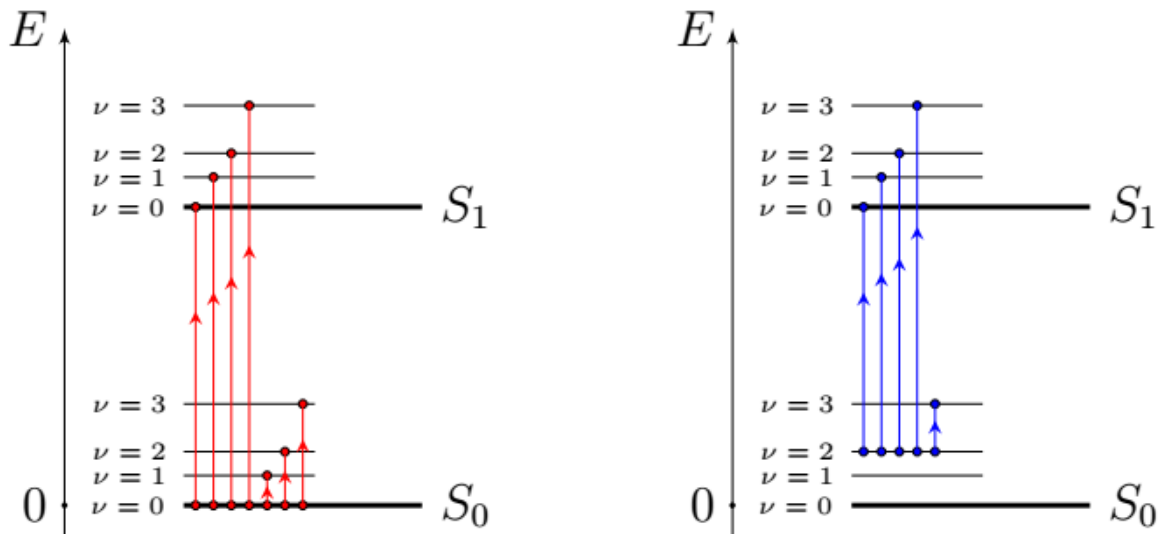


Fig. 5.

The energy of a photon with frequency ω is equal to $\hbar\omega$, where $\hbar$ is Planck's constant. If the molecule is in the $S_0, \nu = 0$ state, and the energy of the photon incident on the molecule coincides with the energy difference between a certain level and the $S_0, \nu = 0$ state, then this photon can be absorbed by the molecule and excite the movement of electrons and vibrations of atoms. In the figure above, colored arrows indicate possible absorption processes for molecules in the $S_0, \nu = 0$ and $S_0, \nu = 2$ states. According to the presented model of the interaction of light with matter, it interacts with photons only when the expression

$$\hbar\omega = E_1 - E_2 \quad (1)$$

выполняется строго и абсолютно точно. В действительности существует множество факторов связанных с более тонким рассмотрением механизма взаимодействия света с веществом, которые допускают поглощение фотонов, для которых (1) выполнено не в точности и график зависимости вероятности $p(\omega)$ поглощения фотона с частотой ω представляет из себя набор пиков конечной ширины. Максимумы этих пиков соответствуют выражению (1). Ширина пиков характеризуется параметром FWHM – полная ширина на полувысоте максимума.

Энергия фотона с частотой ω равна $\hbar\omega$, где $\hbar$ – постоянная Планка. Если молекула находится в состоянии $S_0, \nu = 0$, и энергия налетающего на молекулу фотона совпадает с разницей энергий между некоторым уровнем и состоянием $S_0, \nu = 0$, то этот фотон может быть поглощён молекулой и возбудить движение электронов и колебания атомов. На рисунке выше цветными стрелками указаны возможные процессы поглощения для молекул, находящихся в состояниях $S_0, \nu = 0$ и $S_0, \nu = 2$. Согласно представленной модели взаимодействия света с веществом, оно взаимодействует с фотонами только когда выражение

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Согласно представленной модели взаимодействия света с веществом, оно взаимодействует с фотонами только когда выражение

$$\hbar\omega = E_1 - E_2 \quad (1)$$

выполняется строго и абсолютно точно.

В действительности существует множество факторов связанных с более тонким рассмотрением механизма взаимодействия света с веществом, которые допускают поглощение фотонов, для которых (1) выполнено не в точности и график зависимости вероятности $p(\omega)$ поглощения фотона с частотой ω представляет из себя набор пиков конечной ширины. Максимумы этих пиков соответствуют выражению (1). Ширина пиков характеризуется параметром FWHM – full width at half-maximum.

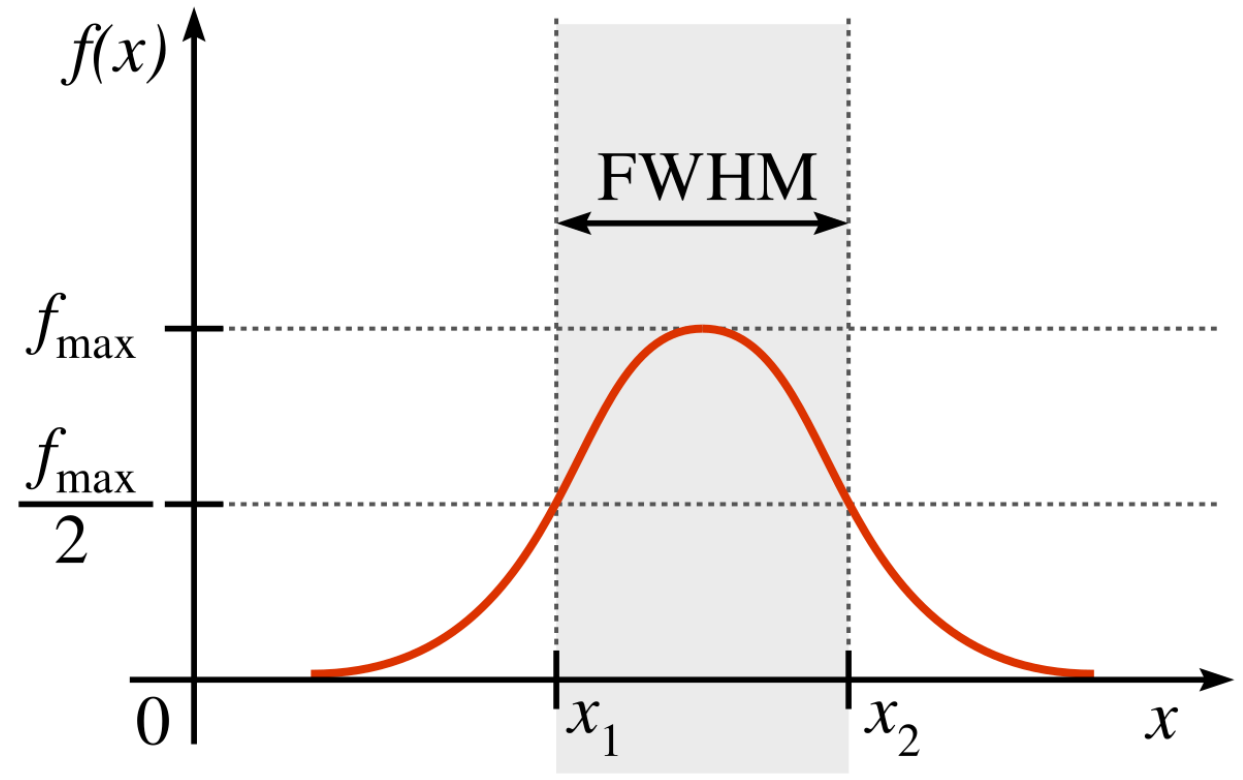


Fig. 6.

An example of the dependence of spectrum broadening on temperature. Y Wang et al. 2021

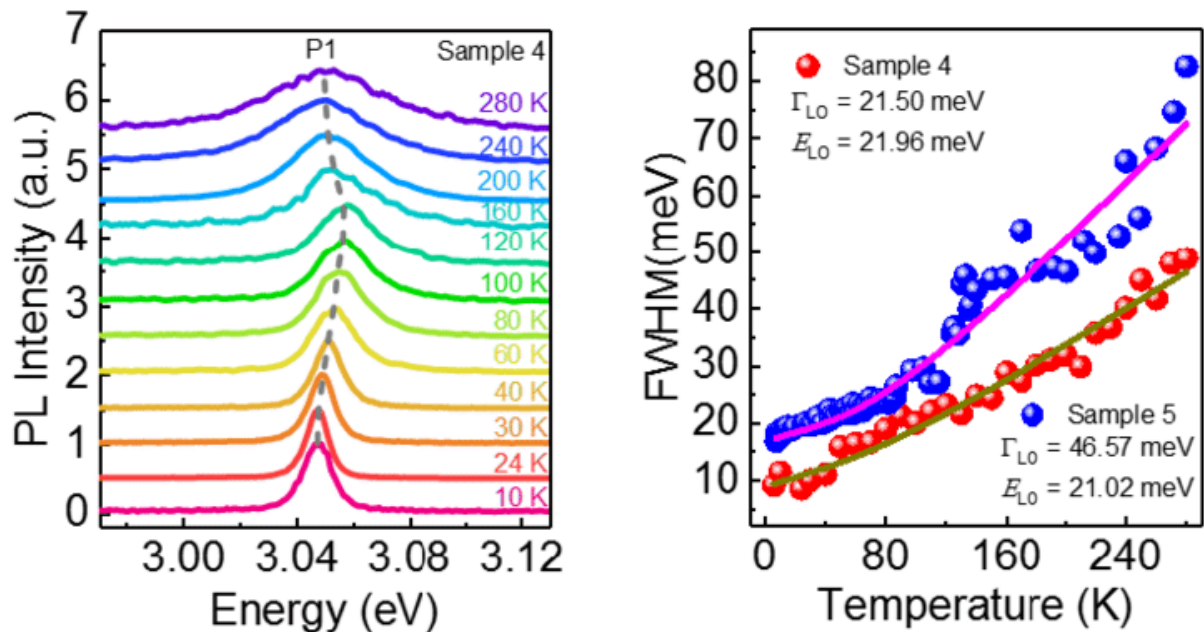


Fig. 7.

States $S_{2,3,\dots}$, etc. will not be considered within the framework of the problem. It is convenient to measure the energies of interaction and motion in atoms in electron volts (eV). $1 \text{ } \vartheta B$ is equal to the energy that an electron acquires by moving between two points in space with a potential difference $1 B$. During the task, concentrations are measured in units of M , $M = 1 \text{ моль/л}$.

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During the task, concentrations are measured in units of M , $M = 1 \text{ моль/л}$.

$\hbar$, Дж $\cdot$ с, c , м/с, e , Кл, N_A , 1/моль $1.05 \cdot 10^{-34}$ $3.00 \cdot 10^8$ $1.60 \cdot 10^{-19}$ $6.02 \cdot 10^{23}$

Part A. Absorption spectrum (0.6 points)

To better understand the described theory, consider a perylene molecule that is not involved in this experiment. The figure below shows the dependence of the probability of absorption p of a photon on its energy $\hbar\omega$ when it enters a perylene molecule in the ground state. The graph shows broadened absorption lines corresponding to the transitions indicated on the graph. The probabilities of different transitions differ, resulting in different heights of absorption peaks.

Absorption spectrum of the ground state of perylene in the visible wavelength range. Berlman, 1971

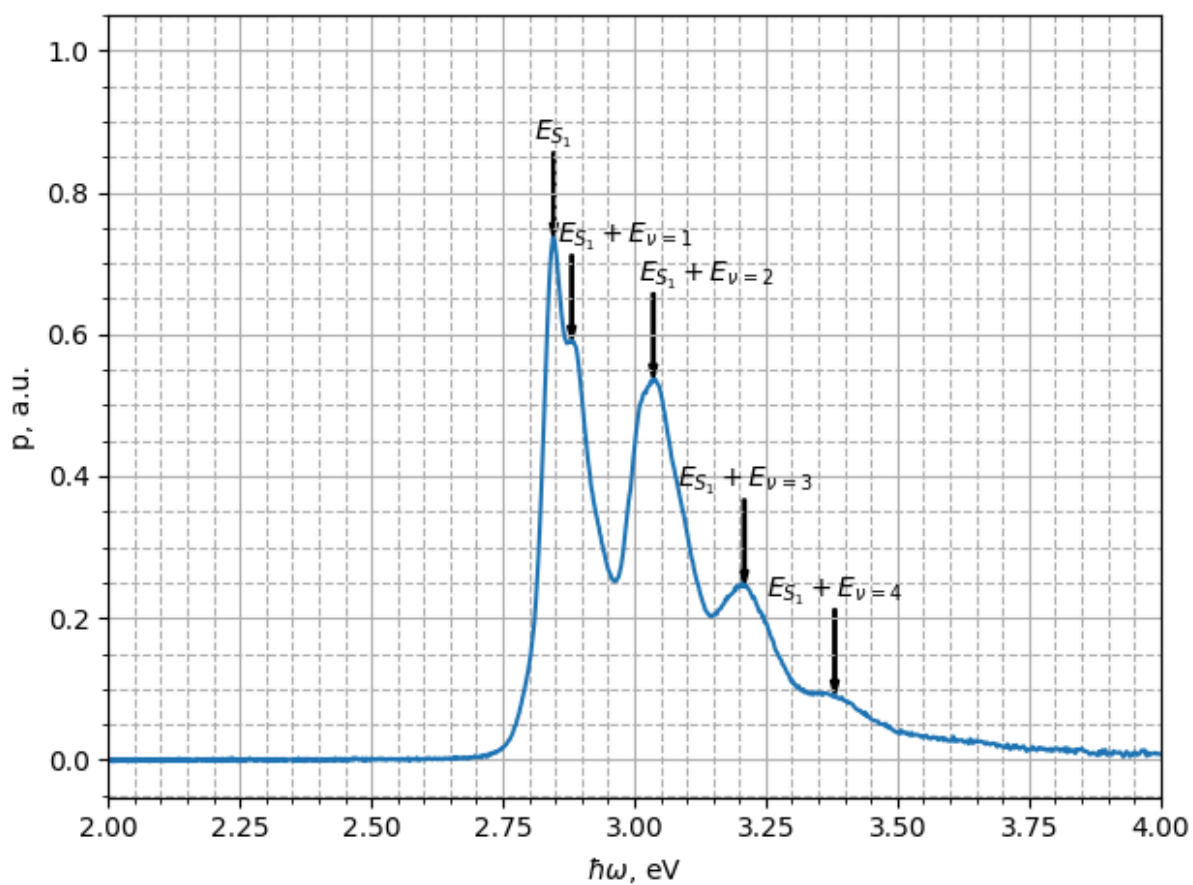


Fig. 8.

A1 Draw a qualitative diagram of Jablonski perylene. Indicate the values of the energies of the states. Take the energy of the ground state equal to zero. **0.30**

The energy of light in the optical range is almost always significantly greater than the energy of the vibrational states of the molecule. Therefore, to study purely vibrational states, absorption spectra are measured in the infrared wavelength range.

Absorption spectrum of the ground state of perylene in the infrared wavelength range. E. Maltseva et al. 2016

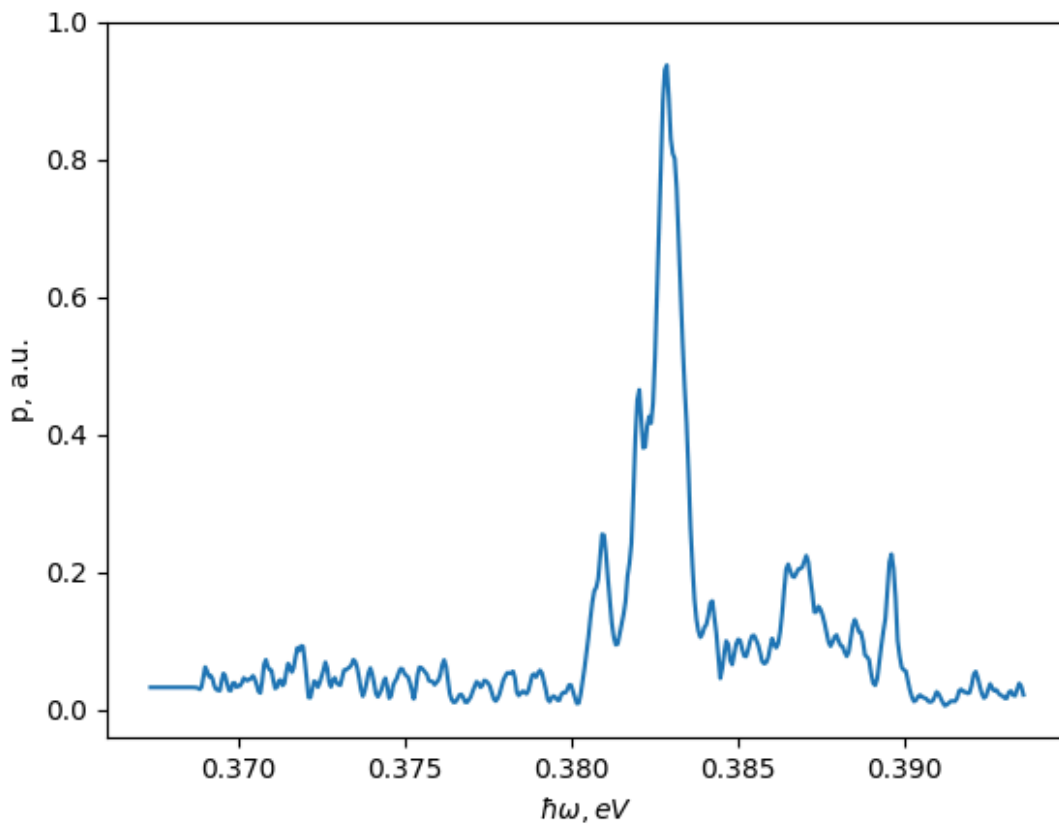


Fig. 9.

- | | | |
|-----------|---|-------------|
| A2 | Indicate the state whose transition corresponds to the absorption maximum in the above graph. | 0.30 |
|-----------|---|-------------|

Part B. Absorption spectroscopy (8.5 points)

- | | | |
|-----------|---|-------------|
| B1 | Determine the focal length of the lens L_3 . | 0.20 |
| B2 | Prepare a solution RhB with a concentration $c = 50 \text{ } \mu\text{M}$. Based on the appearance of the solution, indicate qualitatively which color of light is absorbed most strongly in it. | 0.20 |

We will use the following optical scheme to measure the spectrum of light coming from the source. The slit S allows you to separate the positioning of the image on the CCD ruler and the positioning of the source, since the CCD ruler shows the spectrum of light that appears in the slit.

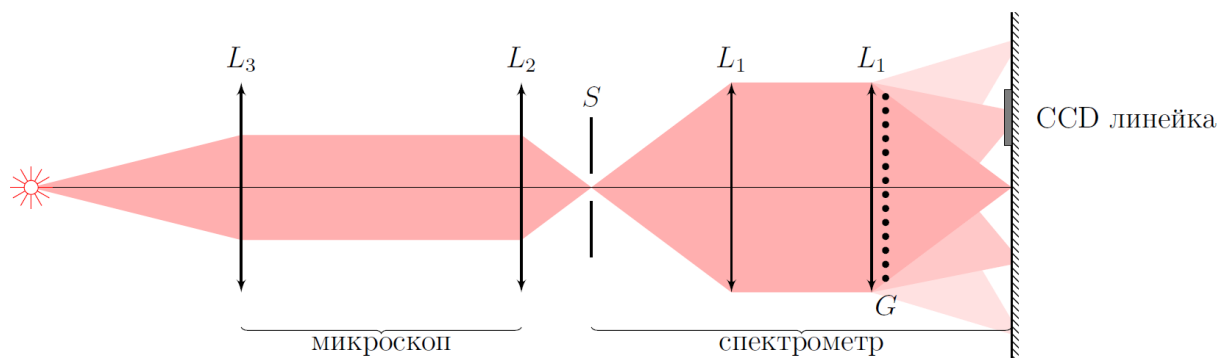


Fig. 10.

Let's find the optimal position of the CCD ruler in our optical design. For this, we will use RGB-LED as a light source with known spectral characteristics. The wavelength $\lambda_{\max}$, which corresponds to the maximum intensity of light emitted by RGB-LED in each operating mode, is indicated in the table below

Color Blue Green Red $\lambda_{\max}$, нм 460 520 620

B3 Adjust the positions of all lenses and the CCD ruler so that the first order of diffraction of visible light falls on the CCD ruler (we are only interested in those positions of the RGB LED when its image in the $L_3 + L_2$ system is focused on the slit). The RGB-LED should be positioned so that the cable is connected to its bottom. After final adjustment of the position of all lenses, measure the emission spectrum $I(\lambda)$ of each of the LEDs. Before any measurement, check that the source image is focused on the slit (the source can be moved). Do not move the lenses and slit until the end of part A, otherwise you will have to do this step again! Save the results of direct measurements as files “R.csv”, “G.csv”, “B.csv” in the “B3” folder on a flash drive. **0.30**

The RGB-LED should be positioned so that the cable is connected to its bottom.

After final adjustment of the position of all lenses, measure the emission spectrum $I(\lambda)$ of each of the LEDs. Before any measurement, check that the source image is focused on the slit (the source can be moved). Do not move the lenses and slit until the end of part A, otherwise you will have to do this step again!

Save the results of direct measurements as files “R.csv”, “G.csv”, “B.csv” in the “B3” folder on a flash drive.

B4 For each RGB-LED color, indicate the pixel number $N_{\max}$ that corresponds to the highest intensity LED emission. **0.30**

A CCD ruler is a sequence of pixels that are located on the “ruler” divisions with a step Δx .

The CCD ruler is a sequence of pixels that are located on the “ruler” divisions with a step of Δx .

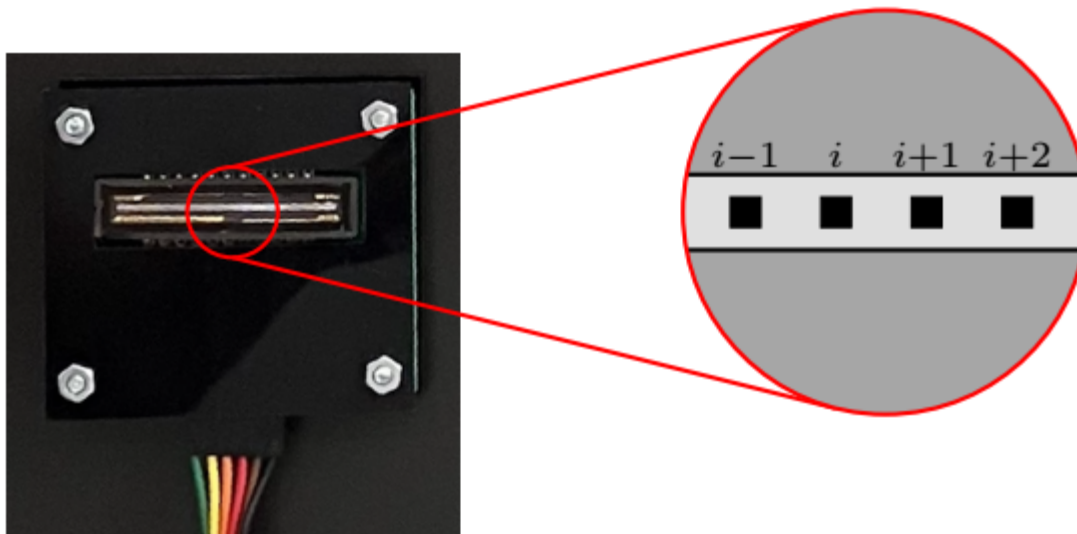


Fig. 11.

B5 You can associate the wavelength λ with the pixel number N using the formula **0.20**

$$\lambda = \text{slope} \cdot N + \text{shift}$$

Indicate the values slope and shift, which follow from the adjustment performed in the previous paragraphs.

In the spectrometer software, you can apply a linear transformation to the horizontal axis for clarity. When saving the file, only the pixel number will be saved, so for any spectra, indicate the adjustment that corresponds to them. Record all adjustments used in the answer sheet table of paragraph B5 and refer to them using their names.–

B6 Find the pixel pitch value Δx . **0.40**

Now the spectral microscope is ready for use

B7 Place a cuvette filled with water on a tripod and focus the LED-CON light on the slit. **0.10**
Qualitatively depict the appearance of the emission spectrum $I_0(\lambda)$ of the LED-CON. Indicate the characteristic wavelengths on the spectrum. Save the results of direct measurements as a file “LED-CON.csv” in the folder “B7”. In your work, specify sensitivity.

When rhodamine is added to water, its refractive index changes slightly, so you can study the passage of light through rhodamine solutions simply by changing cuvettes on a tripod.

B8 Obtain the emission spectrum $I(\lambda)$ of LED-CON light after passing through a solution RhB of concentration $c = 50 \text{ } \mu\text{M}$. Save the results of direct measurements as a file “T-50.csv” in the folder “B8”. Specify sensitivity. **0.40**

B9 Draw a quality view of $I(\lambda)/I_0(\lambda)$. Indicate the wavelength $\lambda_{\text{RhB,max}}$ at which the absorption of light by a rhodamine solution with concentration $c = 50 \text{ } \mu\text{M}$ is maximum. **0.30**

B10 Repeat the measurements of points B7 and B8 for the solution with $c = 50 \text{ } \mu\text{M}$ for 6 other concentrations RhB in the range $[1, 100] \text{ } \mu\text{M}$. Save the results of direct measurements as files “T- c .csv” in the folder “B10”, where c is the concentration of rhodamine in μM . For each file, specify sensitivity. Note: The source spectrum and background may change if you take measurements long enough, so only make test measurements as you prepare solutions. Then take final measurements with clean water ($c = 0 \text{ } \mu\text{M}$) and all 7 prepared solutions. **2.00**

Save the results of direct measurements as files “T- c .csv” in the folder “B10”, where c is the concentration of rhodamine in μM . For each file, specify sensitivity.

Note: The source spectrum and background may change if you take measurements long enough, so only make test measurements as you prepare solutions. Then take final measurements with pure water ($c = 0 \text{ } \mu\text{M}$) and all 7 prepared solutions.

To quantitatively describe the absorption of light in a medium, the molecular absorption cross section $\sigma(\lambda)$ is introduced. The absorption cross section has the dimension of area. If a photon hits the absorption cross section, it is absorbed, and if it doesn't hit, it flies past.

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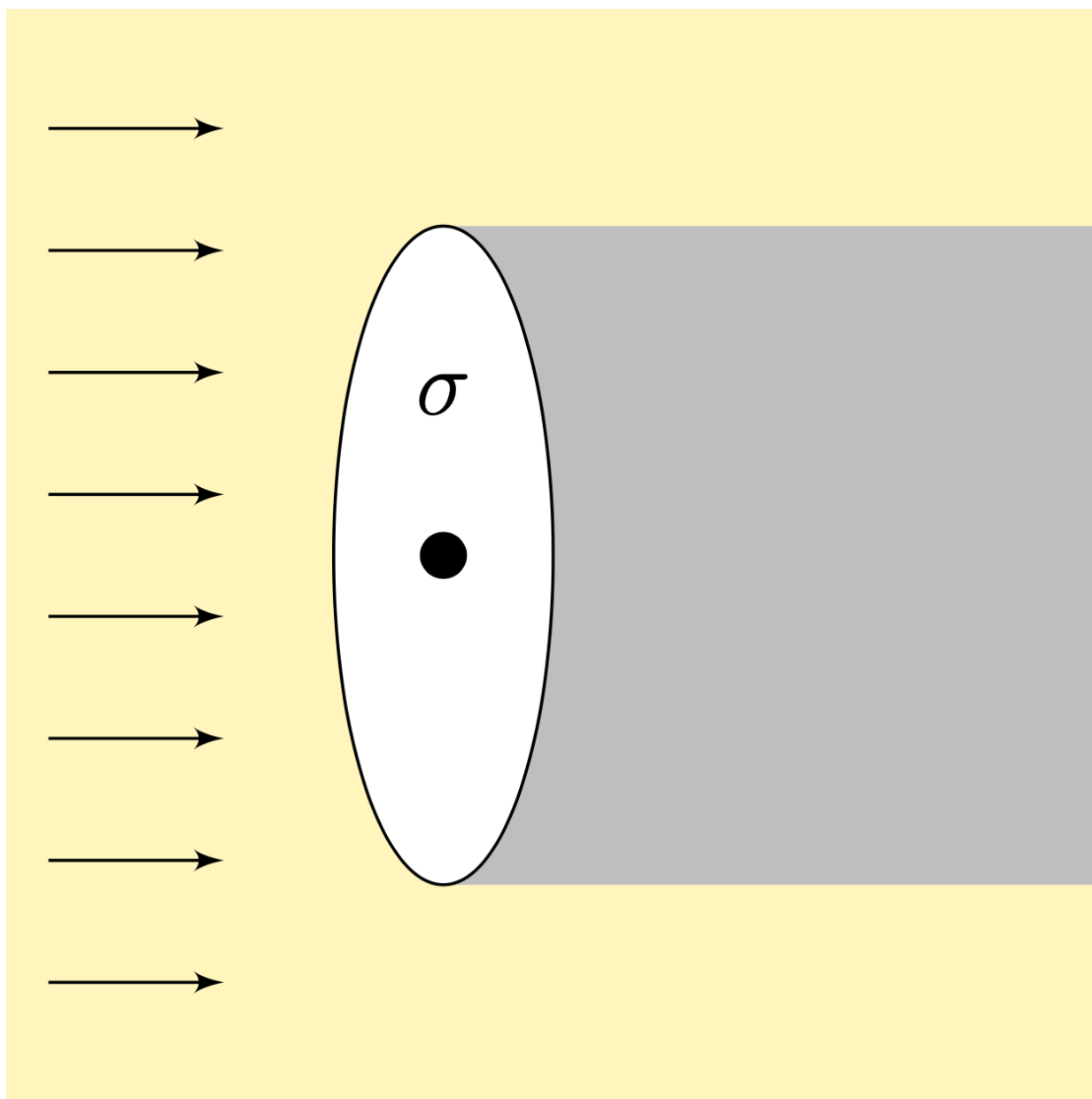


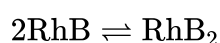
Fig. 12.

- B11** Let monochromatic light with wavelength λ fall on a layer of transparent liquid in which particles are dissolved and absorb the light. The molar concentration of particles is c , the thickness of the transparent liquid layer is L . Determine the ratio I/I_0 of the intensity of light I passed through the solution to the intensity of light I_0 incident on it. Express your answer in terms of σ , c and L . **0.30**

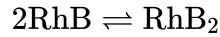
Spectrally complex, LED-CON light can be thought of as a superposition of independent monochromatic waves, each of which interacts with the solution with its own $\sigma(\lambda)$.

- B12** Plot a graph of $\ln(I_0/I)$ for $\lambda = \lambda_{\text{RhB,max}}$ versus c . **0.90**

If the solution RhB has a high concentration, some of the molecules stick together into dimers RhB_2 , which also interact with light. It is convenient to describe such a case as follows: in a solution there are particles of two types that transform into each other according to the following reaction:



In this case, the concentration c that you used in point A12 is not the true concentration of rhodamine monomers RhB. The concentration c corresponds to the concentration of monomers RhB in the case if all dimers in a real solution decomposed into monomers. The ratio c_M/c , where c_M is the real concentration of monomers in the solution, depends nonlinearly on c .



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The ratio c_M/c , where c_M is the real concentration of monomers in the solution, nonlinearly depends on c .

B13 Let monochromatic light with wavelength λ be incident on a layer of transparent liquid in which rhodamine is dissolved so that the concentration of monomers is c_M and the concentration of dimers is c_D . The thickness of the transparent liquid layer is L . The absorption cross section of rhodamine monomers is σ_M , the absorption cross section of rhodamine dimers is σ_D . Determine the ratio I/I_0 of the intensity of light I passed through the solution to the intensity of light I_0 incident on it. Express your answer in terms of σ_M, σ_D , c_M, c_D and L . **0.30**

B14 From the graph view $\ln I_0/I$, indicate the concentrations at which $c_M \simeq c$. **0.20**

B15 Get the formula for converting wavelengths into eV in the form: **0.10**

$$\frac{E}{\text{eV}} = A \cdot \left(\frac{\lambda}{\text{nm}} \right)^\gamma$$

B16 Plot $\sigma_M(E)$ for rhodamine monomers RhB, where E is the energy of the photon interacting with rhodamine. **1.50**

B17 Draw a Jablonski diagram of rhodamine. Indicate the values of the energies of the states. Take the energy of the ground state equal to zero. **0.30**

Below is the absorption spectrum of photons of infrared light from rhodamine, that is, a graph of the dependence of the probability of absorption of a photon on its energy.

Absorption spectrum of the ground state of rhodamine in the infrared wavelength range. Zhou J. et al. 2019.

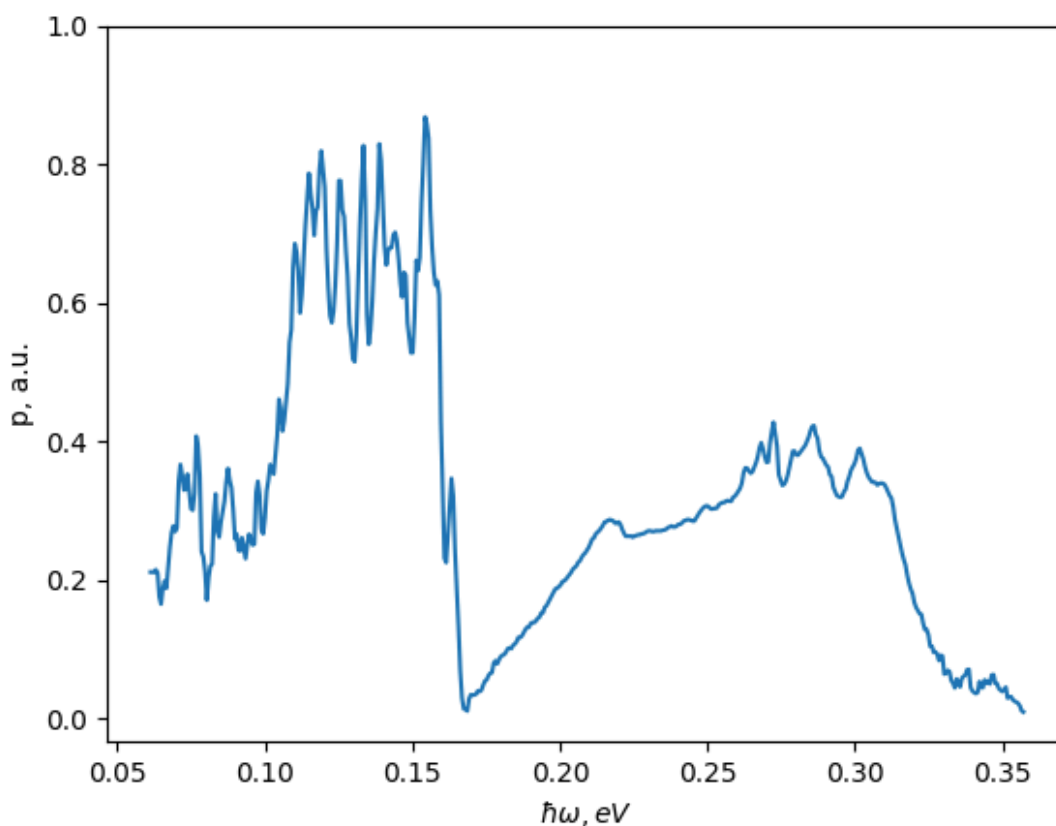


Fig. 13.

B18 Indicate the state whose transition corresponds to the absorption maximum in the above graph. **0.30**

Part C. Fluorescence microscope (10.9 points)

Along with the process of absorption of light by a substance, the emission of light is possible if the molecule is in an excited state. In this case, again $\Delta E = \hbar\omega$, only now ω is the frequency of the emitted photon. This process is called spontaneous emission.

All possible processes of photon emission when the molecule is in the $S_1, \nu = 0$ state (left), in the $S_1, \nu = 1$ state (right).

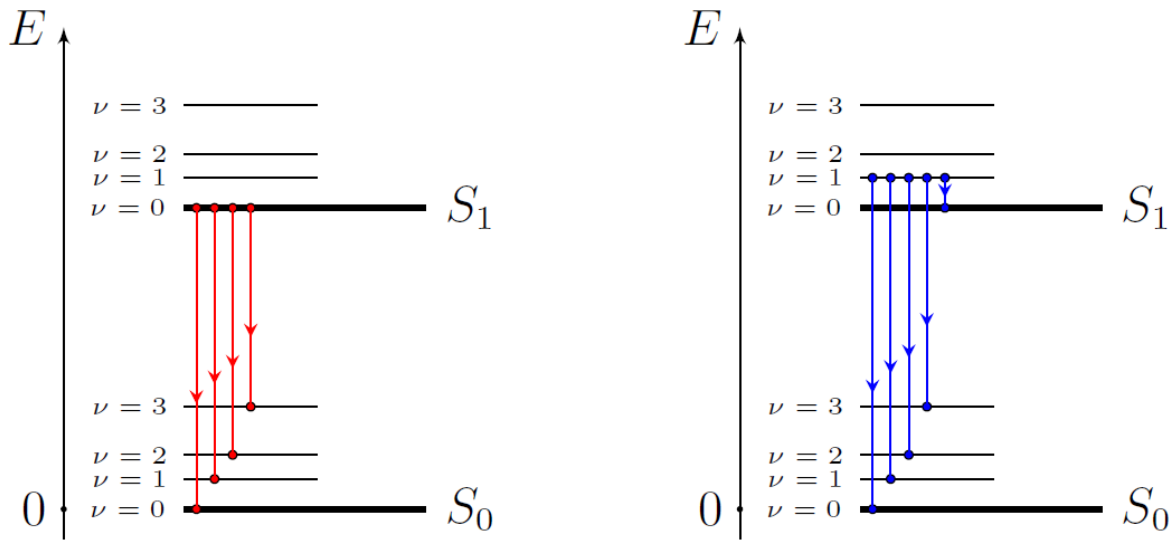


Fig. 14.

In fact, the emission of photons from states $S_{1,\nu \neq 0}$ is not observed in the experiment. This is explained by non-radiative relaxation, associated with the generation of mechanical waves with almost optical frequency (phonons) in the liquid surrounding the molecule. The simplest way to transfer a molecule to the excited state $S_{1,\nu}$ is to illuminate it with laser radiation with photon energy $\hbar\omega_0 > E_{S_1}$. In this case, a process occurs consisting of three stages: Absorption of an exciting photon with energy $\hbar\omega_0 > E_{S_1}$. Non-radiative relaxation from levels $E_{S_1,\nu \neq 0}$ to $E_{S_1,0}$. Emission of a photon with energy $\hbar\omega < E_{S_1}$. During emission, the photon does not necessarily reach the $E_{S_0,\nu=0}$ level. When entering vibrational sublevels, the molecule relaxes nonradiatively to the ground state according to the same mechanism as described above. The entire process described is called fluorescence. Considering the processes of fluorescence and optical absorption together allows us to obtain a much more complete picture of the Jablonski diagram of a molecule.

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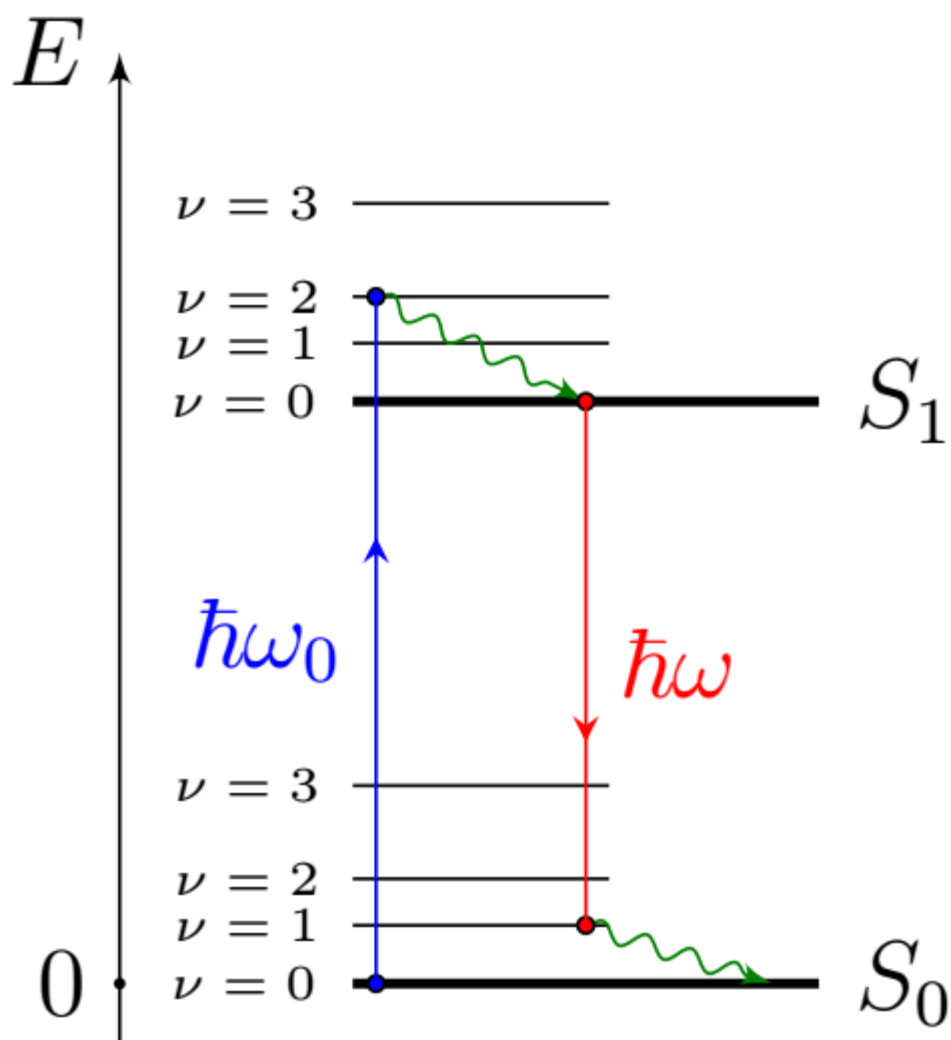


Fig. 15.

The interpretation of the perylene absorption peaks indicated at the very beginning of Part A can be proven based on a comparison of the absorption and emission spectra

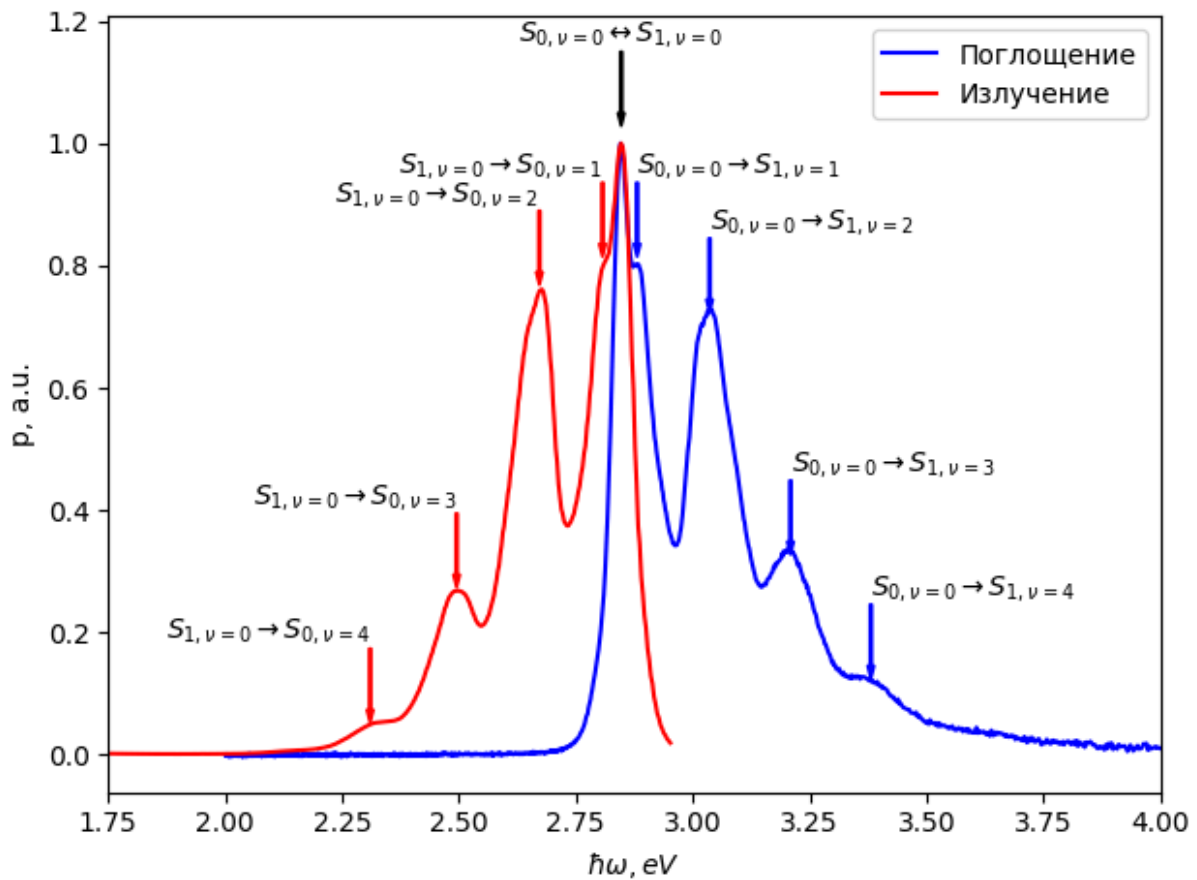


Fig. 16.

To excite fluorescence we will use laser beams. Their wavelengths are: Color Blue Green Red λ , μm 405 532 650

To excite fluorescence we will use laser beams. Their wavelengths:

Next, you may need a spacer - a small plastic plate. It is designed to adjust the installation height of the illuminator in the holder. You can install it in the light holder as shown in the figure, and then install the required light in the same guides.

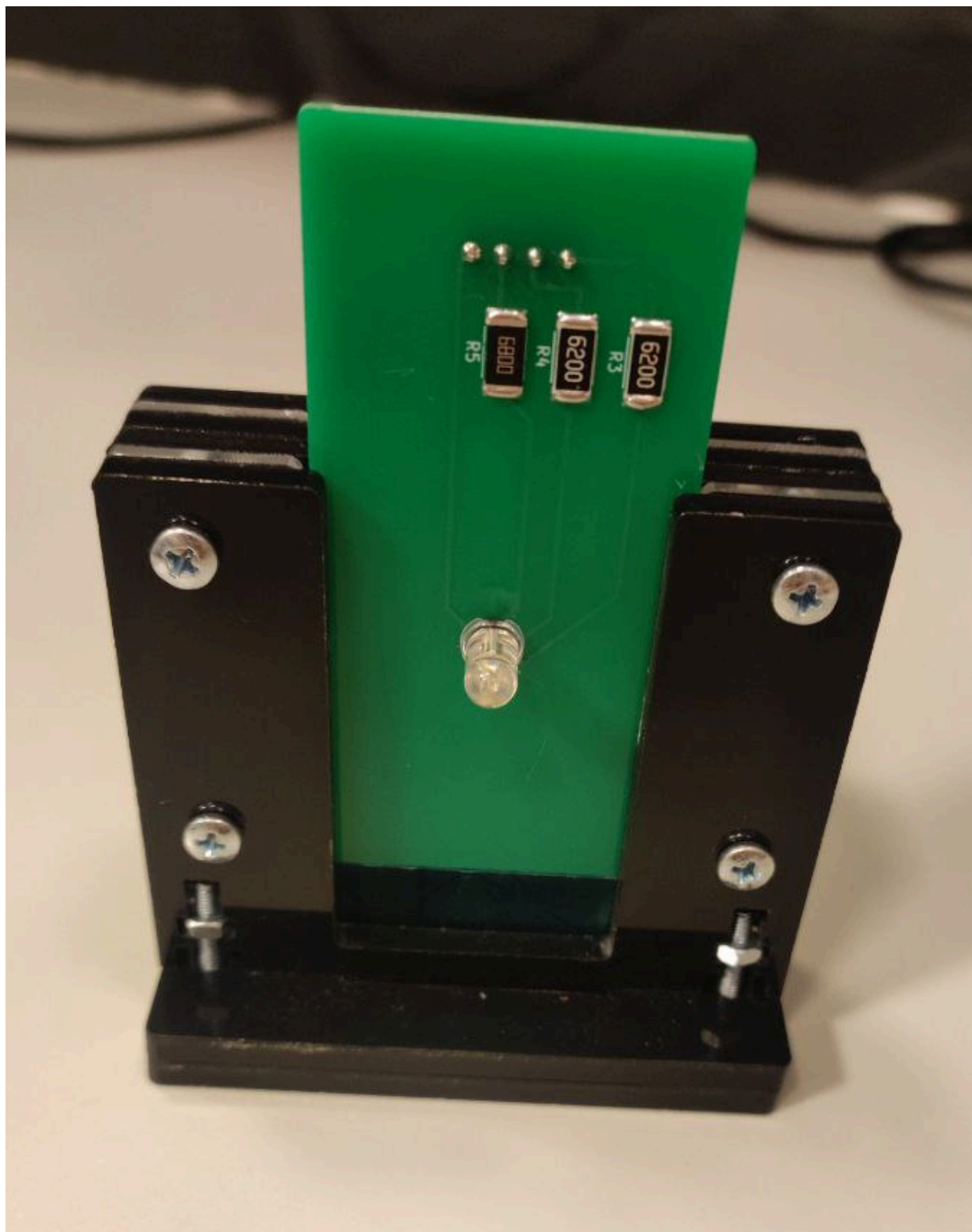


Fig. 17.

C1 Shine the light on the cuvette containing the rhodamine solution with each laser. Describe the observed optical phenomena (absorption and emission) and explain them using the Jablonski diagram you obtained at the end of the previous part. **0.30**

The highest absorption and emission in perylene corresponds to the transition between states with $\nu = 0$, however, in most fluorescent molecules this transition occurs with a low probability, and the maximum absorption corresponds to the transition with $S_{0,\nu=0} \rightarrow S_{1,\nu=1}$. In this case, the maximum emission corresponds to the transition $S_{1,\nu=0} \rightarrow S_{0,\nu=0}$. This leads to the fact that the absorption and emission maxima

do not coincide and are shifted relative to each other. This shift is called the Stokes shift. The presence of a Stokes shift in rhodamine shows that the Jablonski diagram we obtained in the previous part may be slightly incorrect, since we may have incorrectly identified the transitions.

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In this part, we modify the optical design to collect greater light intensity on the CCD line. When changing the circuit, you should not change the position of the lenses inside the spectrometer.

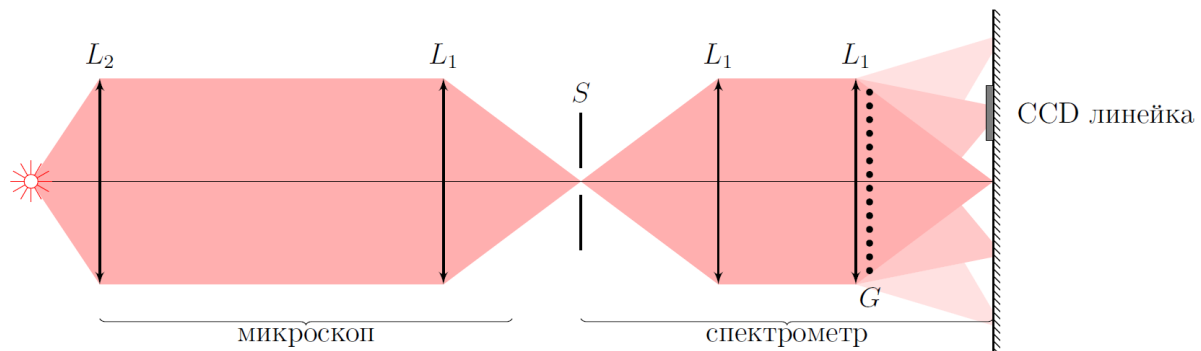


Fig. 18.

Adjust the position of lenses L_2 and L_1 using the inverted RGB-LED.

- | | | |
|-----------|---|-------------|
| C2 | Again measure the emission spectrum $I(\lambda)$ of each of the LED-RGB LEDs. Before any measurement, check that the source image is focused on the slit (the source can be moved). Do not move the lenses and slit until the end of part B, otherwise you will have to do this step again! Save the results of direct measurements as files “R.csv”, “G.csv”, “B.csv” in the “C2” folder on a flash drive. | 0.30 |
|-----------|---|-------------|

Save the results of direct measurements as files “R.csv”, “G.csv”, “B.csv” in the “C2” folder on a flash drive.

For further measurements, use only the violet and green lasers, which shine vertically into the open cell. When monochromatic radiation with sufficient photon energy enters a fluorescent medium, it is gradually absorbed as it passes through the solution and transfers the molecules to an excited state. In the excited state $S_{1,\nu=0}$ of a rhodamine molecule there is an average time $\tau_0 \approx 1.2 \text{ ns}$, which is called the lifetime of the excited state.

- | | | |
|-----------|--|-------------|
| C3 | Let on average one photon hit a rhodamine molecule located in a laser beam during time τ . Express τ in terms of the absorption cross section of the molecule σ , the intensity of the laser radiation I_0 and the frequency ω_0 of the laser radiation. | 0.20 |
|-----------|--|-------------|

- | | | |
|-----------|--|-------------|
| C4 | Estimate the intensity I_0 of light at the focus of lens L_3 if light from a blue laser falls on it. Laser radiation power $P_0 = 5 \text{ mW}$, beam diameter $d_0 = 2 \text{ mm}$. Also estimate the value of τ . | 0.90 |
|-----------|--|-------------|

- C5** What is the power P of fluorescent radiation per unit volume of a solution with rhodamine exposed to a laser beam with intensity I_0 and frequency ω_0 ? Consider the known absorption cross section σ , the excited state lifetime $\tau_0 \ll \tau$ and the molar concentration of rhodamine c . Also consider that fluorescence occurs in a narrow frequency range close to ω . If you haven't done this, consider it known that $P \propto I_0 c$. **0.40**

Let us obtain the fluorescence emission spectra of the spectrometer microscope. To do this, you need to VERY carefully select the position of the focused vertical laser beam so that fluorescence occurs at the focus of the objective lens L_2 .

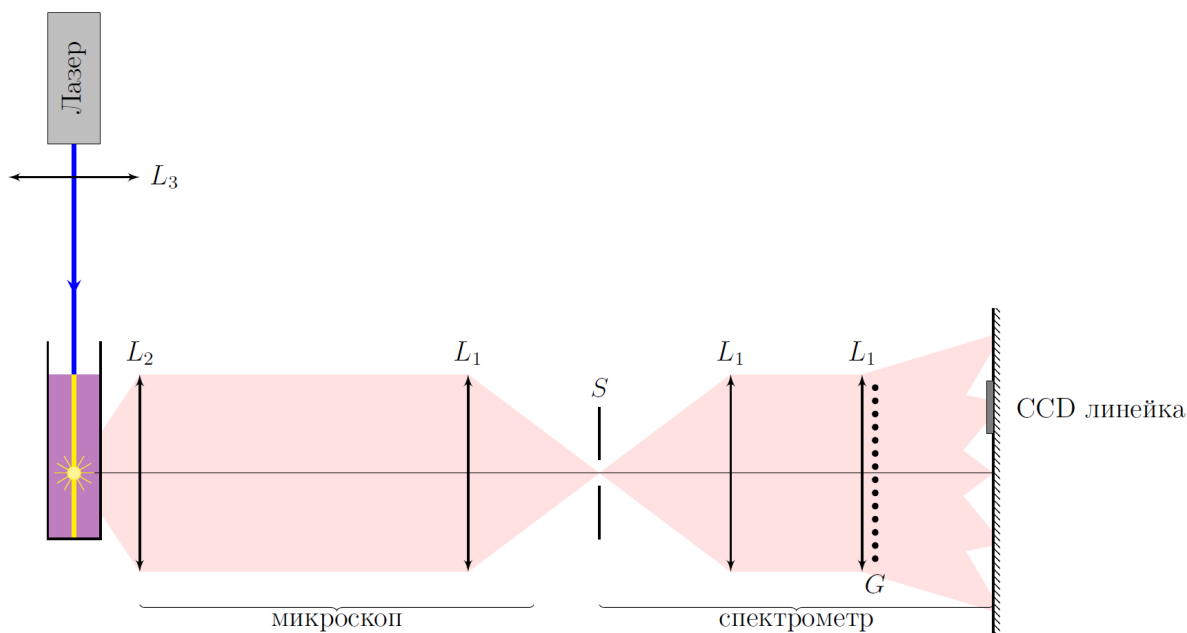


Fig. 19.

- C6** Obtain the fluorescence spectrum of rhodamine when excited by a violet laser in a solution with a concentration of 100 μM . Draw a qualitative graph of the fluorescence spectrum of $I_{405}(\lambda)$. Indicate the wavelength $\lambda_{405, \text{max}}$ at which fluorescence emission is maximum. Save the results of direct measurements as a file "405.csv" in the "C6" folder on a flash drive. Do not change any parts of the installation other than the cuvettes during the following steps. **0.50**

Draw a qualitative graph of the fluorescence spectrum of $I_{405}(\lambda)$. Indicate the wavelength $\lambda_{405, \text{max}}$ at which fluorescence emission is maximum.

Save the results of direct measurements as a file "405.csv" in the "C6" folder on a flash drive.

Do not change any parts of the installation other than the cuvettes during the following steps.

- C7** Remove most of the rhodamine solution from the cuvette using a 3 ml syringe. Record the volume V_0 of the solution inside the syringe. By gradually adding the solution to the cuvette, study the dependence of the fluorescence intensity $I_{405}(\lambda = \lambda_{405, \text{max}})$ on the volume of liquid V inside the syringe. Save the results of direct measurements as files "405-V.csv" in the "C7" folder on a flash drive, where V is the volume of liquid in the syringe. **2.30**

C8	Plot a linearized graph of I versus V . Determine the absorption cross section σ_{405} of rhodamine molecules at the blue laser wavelength.	1.50
C9	For four other concentrations of rhodamine c , measure the fluorescence spectrum of $I_{405}(\lambda)$. Save the results of direct measurements as a file “405- c .csv” in the “C9” folder on a flash drive, where c is the concentration of rhodamine in μM . For each file, specify sensitivity.	2.00
C10	Plot a linearized graph of $I_{405}(\lambda_{405,\text{max}})$ versus c .	1.20
C11	Obtain the fluorescence spectrum of rhodamine under green laser excitation. Draw a qualitative graph of the fluorescence spectrum of $I_{532}(\lambda)$. Save the results of direct measurements as a file “532.csv” in the “C11” folder on a flash drive.	0.20

To compare the absorption spectra measured in part A and in part B, we compare two dependences: Normalized absorption spectrum: $A(E) = \sigma_{\text{M}}(E)/\sigma_{\text{M}}(E_{\text{max}})$, where E_{max} is the energy of photons that are most strongly absorbed by rhodamine Normalized fluorescence spectrum: $F(E) = I_{405}(E)/I_{405}(E_{405,\text{max}})$, where $E_{405,\text{max}}$ is the energy of photons, the emission of which is strongest during fluorescence.

To compare the absorption spectra measured in part A and part B, let us compare two dependences

C12	On one high-quality graph, draw the graphs $A(E)$ and $F(E)$. Indicate the characteristic energies of all features.	0.80
C13	Draw a qualitative Jablonski diagram of rhodamine. Indicate the values of the energies of the states. Take the energy of the ground state equal to zero.	0.50

Source: <https://pho.rs/p/4541>